Lecture 6

Damped Oscillations: Equation

$$\frac{d^2y}{dt^2} + 2\lambda\frac{dy}{dt} + \omega^2 y = 0 \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where $2\lambda = \frac{b}{m}$ and $\omega^2 = \frac{a}{m}$

Equation (1) which is a homogeneous linear type differential equation of the second order, must have at least one solution of the form y=Ae^{kt},where A and k are both arbitrary constants.

Let, y=Ae^{kt}, Differentiating y with respect to time, we get,

$\frac{dy}{dt}= kAe^{kt}$ and $\frac{d^2y}{dt^2}=k^2\mathrm{A}e^{kt}$

Substituting these values in equation (1); we have

$k^2\mathrm{A}e^{kt} + 2\lambda\, kAe^{kt} + \omega^2\, Ae^{kt}=0$

Dividing throughout by Ae^{kt}, we have

$k^2 + 2\lambda k + \omega^2=0$ ………………………………………….. (2)

Equation (2) is clearly a quadratic equation in k, the solution of which is;

k= $-\lambda \pm \sqrt{\lambda^2 - \omega^2}$ [Quadratic formula,

$ax^2 + bx + c = 0, then\ x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$]

the differential equation (1) is, therefore, satisfied by two values of y, viz,

$$y = Ae^{(-\lambda+\sqrt{\lambda^2-\omega^2})t}$$

And

$$y = Ae^{(-\lambda-\sqrt{\lambda^2-\omega^2})t}$$

The equation being a linear one, the linear sum of the two linearly independent solution of the equation is the most general solution. Thus the general solution is,

$y = A_1 e^{(-\lambda+\sqrt{\lambda^2-\omega^2})t} + A_2 e^{(-\lambda-\sqrt{\lambda^2-\omega^2})t}$ .... (3)

Where $A_1 and\ A_2$ are two arbitrary constants.

The values of the arbitrary constants $A_1$and A_2 may be determined as follows:

$$\frac{dy}{dt} = \left(-\lambda + \sqrt{\lambda^2 - \omega^2}\right) A_1 e^{(-\lambda+\sqrt{\lambda^2-\omega^2})t} +$$
$$\left(-\lambda - \sqrt{\lambda^2 - \omega^2}\right) A_2 e^{(-\lambda-\sqrt{\lambda^2-\omega^2})t} \text{............ (4)}$$

Let the maximum value of the displacement y be $y_{max} = a_o$,say at time t=0. Then from equation (3);

$y_{max} = a_o = A_1 + A_2$... (5)

Again, the velocity is zero at maximum displacement, i.e., $\frac{dy}{dt} = 0$ at t=0.

Hence from equation (4), we have

$\left(-\lambda + \sqrt{\lambda^2 - \omega^2}\right)A_1 + \left(-\lambda - \sqrt{\lambda^2 - \omega^2}\right)A_2=0$

or, $-\lambda(A_1 + A_2)+ \sqrt{\lambda^2 - \omega^2}(A_1 - A_2)=0$

or, $\sqrt{\lambda^2 - \omega^2}(A_1 - A_2)= \lambda(A_1 + A_2)= \lambda a_o$

or, $(A_1 - A_2)=\frac{\lambda a_o}{\sqrt{\lambda^2-\omega^2}}$... (6)

Adding equation (5) & (6), we have

$2A_1= a_o + \frac{\lambda a_o}{\sqrt{\lambda^2-\omega^2}}$

or, $A_1 = \frac{1}{2}(a_o + \frac{\lambda a_o}{\sqrt{\lambda^2-\omega^2}}) = \frac{1}{2}a_o(1 + \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})$

Now subtracting equation (6) from equation (5), we have

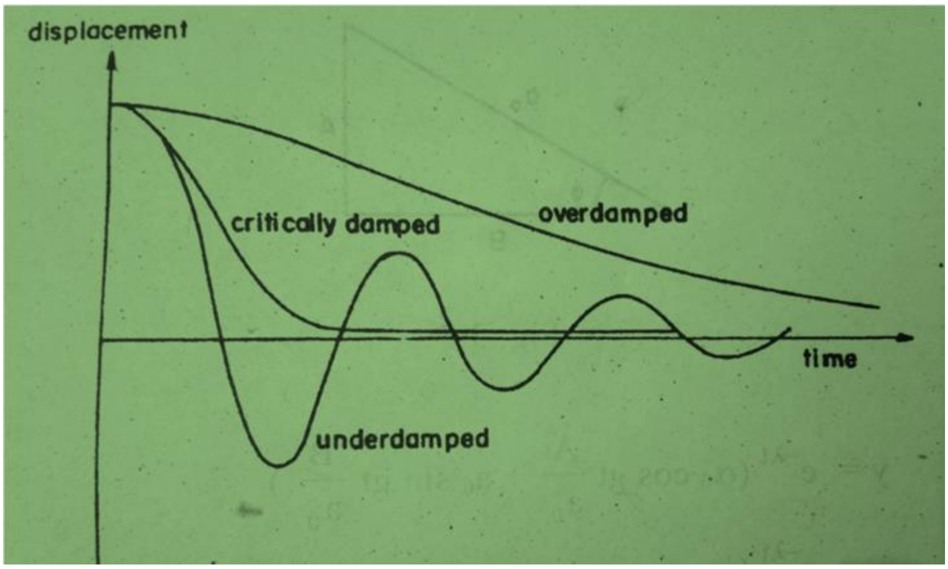
$A_2 = \frac{1}{2}(a_o - \frac{\lambda a_o}{\sqrt{\lambda^2-\omega^2}}) = \frac{1}{2}a_o(1 - \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})$

Substituting these values of A_1 and A_2 in equation (3), we have

$$y = \frac{1}{2}a_o(1 + \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})e^{(-\lambda+\sqrt{\lambda^2-\omega^2})t} + \frac{1}{2}a_o(1 - \frac{\lambda}{\sqrt{\lambda^2-\omega^2}}) e^{(-\lambda-\sqrt{\lambda^2-\omega^2})t}$$

……………………(7)

Three important cases now arise:



- $\boldsymbol{\lambda^2 > \omega^2}$**:** Damping in large; $\sqrt{\lambda^2-\omega^2}$ is clearly a real quantity with a positive value, less than λ.Thus each of the two terms on the right hand side of equation (7) has an exponential term with a negative power and hence each decreases exponentially with time. In this case the particle does not vibrate. There is thus no oscillation and the motion is, therefore, called overdamped.
- $\boldsymbol{\lambda^2 = \omega^2}$**:** In this case, $\sqrt{\lambda^2-\omega^2}$ is obviously equal to zero, so that each of the two terms on the right hand side of equation (7) becomes infinite and the solution breaks down.

Let us, however, consider that λ^2 is not quite equal to ω^2, but very nearly so, so that $\sqrt{\lambda^2 - \omega^2}$= h, a very small quantity but not zero. Then equation (3) becomes,

$$y = A_1 e^{(-\lambda+h)t} + A_2 e^{(-\lambda-h)t}$$

$$= e^{-\lambda t}(A_1 e^{ht} + A_2 e^{-ht})$$

Or, $y = e^{-\lambda t}[A_1(1 + ht + \frac{h^2t^2}{2!} + \frac{h^3t^3}{3!} + \cdots) + A_2(1 - ht + \frac{h^2t^2}{2!} - \frac{h^3t^3}{3!} + \cdots)]$

Neglecting terms containing second and higher powers of h, we get

$y = e^{-\lambda t}[A_1(1 + ht) + A_2(1 - ht)]$

or, $y = e^{-\lambda t}[(A_1 + A_2) + (A_1 - A_2)ht]$............................... (8)

Putting $(A_1 + A_2) = M$ and $(A_1 - A_2)h$= N;

$y = e^{-\lambda t}[M + Nt]$

Recalling that $y = y_{max} = a_o = A_1 + A_2 = M$

And , $\frac{dy}{dt} = 0$ at t=0.

$(A_1 - A_2) = \frac{\lambda a_o}{\sqrt{\lambda^2 - \omega^2}} = \frac{\lambda a_o}{h}$

Or, $(A_1 - A_2)h = \lambda a_o = N$

Equation (8), therefore, becomes

$y = e^{-\lambda t}[a_o + \lambda a_o t]$

$= e^{-\lambda t} a_o[1 + \lambda t]$.. (9)

The displacement of the oscillator first increases but as t increase the exponential factor $e^{-\lambda t}$ becomes more important and the displacement decreases rapidly reaching the value zero for a finite value of t. The oscillator just ceases to oscillate and its motion just becomes aperiodic or non-oscillatory. This is called the case of critical damping, the necessary condition for which is that λ^2 should be almost but not quite equal to ω^2.

- $\boldsymbol{\lambda^2 < \omega^2}$**:** the quantity $\sqrt{\lambda^2 - \omega^2}$ is clearly imaginary, say equal to ig where i=$\sqrt{-1}$ and g = $\sqrt{\omega^2 - \lambda^2}$ is a real quantity.

Equation (3) becomes

$y = A_1 e^{(-\lambda+ig)t} + A_2 e^{(-\lambda-ig)t}$

$= e^{-\lambda t}(A_1 e^{igt} + A_2 e^{-igt})$

$= e^{-\lambda t}[A_1(\cos gt + i\, singt) + A_2(\cos gt - i \sin gt)]$

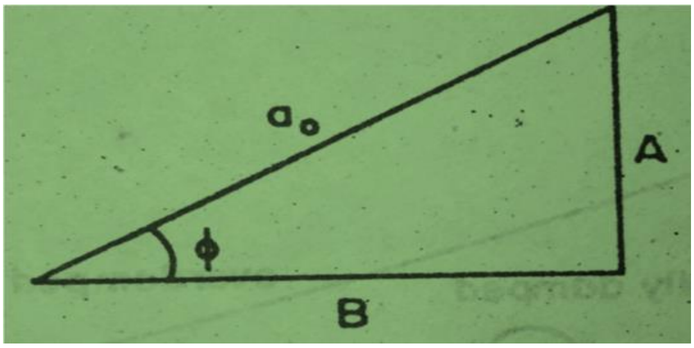
$= e^{-\lambda t}[(A_1 + A_2)\cos gt + i\,(A_1 - A_2)\sin gt]$

Putting $(A_1 + A_2) = A$ and

$i(A_1 - A_2) = B,$ we have

y = $e^{-\lambda t}[A \cos gt + B \sin gt]$

If A, B and a_o be related as shown in figure



y = $e^{-\lambda t}[a_o \cos gt \frac{A}{a_o} + a_o \sin gt \frac{B}{a_o}]$

$= e^{-\lambda t}[a_o \cos gt\ sin\phi + a_o \sin gt\ \ cos\phi]$

$= e^{-\lambda t} a_o \sin(gt + \phi)$

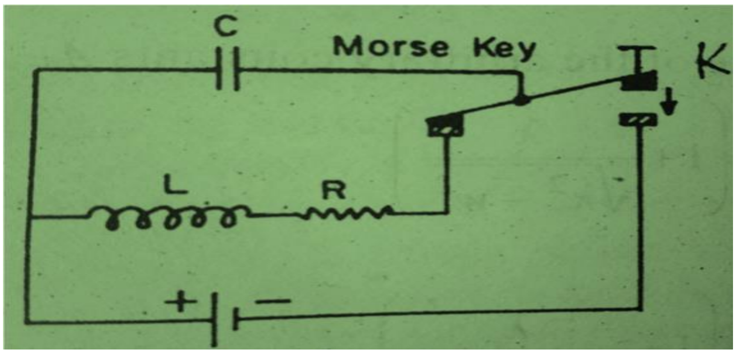
$= a_o e^{-\lambda t} \sin(gt + \phi)$……………………………………………..(10)

Equation (10) is the equation of a damped harmonic oscillator with amplitude $a_o e^{-\lambda t}$ and frequency g/2π. It is so called because the sine term in the equation suggests the oscillatory character of the motion and the exponential term, the gradual damping out of the oscillations. Thus although the amplitude of an underdamped harmonic oscillator decreases exponentially with time, it does perform a sort of oscillatory motion. The motion does not, of course, repeat itself and is thus not periodic in the usual sense of the term.

Damped Oscillations: LRC Circuit

An electrical circuit which contains inductance (L), capacitance (C) and resistance (R) renders an excellent example of damped harmonic oscillations.

Figure shows a circuit containing a resistance, an inductance ad a capacitance connected in series. A battery can be included into the circuit through the key K. When K is pressed, the capacitor is charged. On releasing K, the battery is thrown out of the circuit and the capacitance discharges through the resistance and the inductance.



Suppose that at a given instant t, the charge on the capacitor is Q and the current flowing through the circuit is i. Then the induced emf. Across the inductance L is L $\frac{di}{dt}$ and the potential difference across the resistance R is Ri. Since there is no external emf in the circuit now (the battery being disconnected), then according to Kirchoff's law,

$$Ri+\frac{Q}{C}=-L\frac{di}{dt}$$

Or, Ri+$\frac{Q}{C}$+L$\frac{di}{dt}$=0

Or, , L$\frac{di}{dt}$+Ri+$\frac{Q}{C}$=0……………………………………(1)

But i=$\frac{dQ}{dt}$; therefore equation (1) becomes

$$L\frac{d^2Q}{dt^2} + R\ \frac{dQ}{dt} + \frac{Q}{C} = 0$$

Or, $\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{Q}{LC} = 0$………………………………………… (2)

Equation (2) is very similar to the equation of the differential equation of damped harmonic oscillation, with the difference that the displacement y is replaced by charge Q, 2λ by $\frac{R}{L}$ and ω^2 by $\frac{1}{LC}$.

The solution of equation (2);

$$Q = A_1e^{(-\lambda+\sqrt{\lambda^2-\omega^2})t} + A_2e^{(-\lambda-\sqrt{\lambda^2-\omega^2})t}$$

Where the value of the arbitrary constants A_1 and A_2 are;

$A_1 = \frac{1}{2}Q_o(1 + \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})$

And $A_2 = \frac{1}{2}Q_o(1 - \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})$

Thus $$Q = \frac{1}{2}Q_o(1 + \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})e^{(-\lambda+\sqrt{\lambda^2-\omega^2})t} + \frac{1}{2}Q_o(1 - \frac{\lambda}{\sqrt{\lambda^2-\omega^2}})e^{(-\lambda-\sqrt{\lambda^2-\omega^2})t} \quad \text{…………(3)}$$

Again the following three cases arise:

- When the damping is large, i.e.,
  $\lambda^2 > \omega^2$(or, $\frac{R^2}{4L^2} > \frac{1}{LC}$ or, $R^2 > \frac{4L}{C}$)

  Thus the discharge of the capacitor is non-oscillatory or damping.
- When the damping is critical i.e.,

$\lambda^2 = \omega^2$(or, $R^2 = \frac{4L}{C}$). Here the discharge of the condenser just becomes aperiodic and dies down in the shortest possible time.

- When the damping is small i.e., $\lambda^2 < \omega^2$(or, $R^2 < \frac{4L}{C}$). Thus equation for he discharge of charge takes the form

  Q = $Q_o e^{-\lambda t} \sin(gt + \phi)$.. (4)

  Differentiating equation (4) with respect to time, we have

  $$\frac{dQ}{dt} = \frac{dQ_o}{dt} e^{-\lambda t} \sin(gt + \phi)$$

  $i = i_o e^{-\lambda t} \sin(gt + \phi)$; where i_o the maximum value or the amplitude of the current.

  The frequency of oscillation is

  $$n = \frac{g}{2\pi} = \frac{\sqrt{\omega^2 - \lambda^2}}{2\pi} = \frac{1}{2\pi}\sqrt{\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)}$$

  In the absence of R i.e., if R=0, the frequency of oscillation becomes

  $$n = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$$

  The same as that of an LC circuit, with no damping of the oscillation.

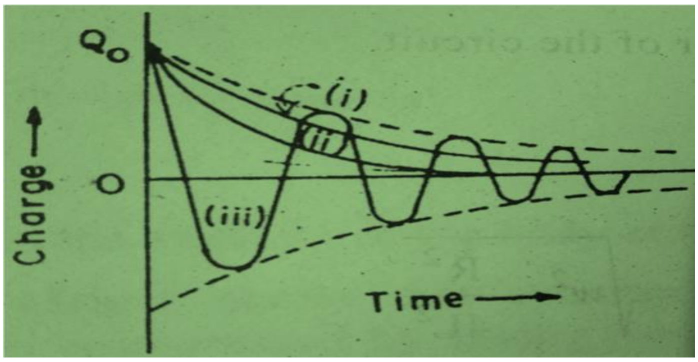
  The quality factor of the circuit,

  $$Q = \frac{g}{2\lambda} = \frac{Lg}{R}$$

  Where g= $\sqrt{\omega^2 - \lambda^2} = \sqrt{\left(\omega^2 - \frac{R^2}{4L^2}\right)}$

  Since R is small, g$\simeq \omega$

  And therefore, the quality factor, Q=$\frac{L\omega}{R}$



□ **Underdamped**:

- **Description**: Weak damping; system oscillates with decreasing amplitude.
- **Behavior**: Multiple oscillations before coming to rest..

□ **Critically Damped**:

- **Description**: Optimal damping; system returns to equilibrium as quickly as possible without oscillating.
- **Behavior**: No oscillations, fastest return to equilibrium.

□ **Overdamped**:

- **Description**: Strong damping; system returns to equilibrium slowly without oscillating.
- **Behavior**: No oscillations, slower return to equilibrium than critically damped.